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A Reimplementation and Upgradation of ProsePose with SAM 3D Body

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ABSTRACT Estimating accurate 3D human pose from a single image in multi-person scenarios is fundamentally complicated by occlusion, self-contact, and depth ambiguity. Existing Human Mesh Recovery (HMR) models, including the BEV and HMR2 regressors used in the original ProsePose paper, process each person independently and consequently produce meshes that either float apart at genuine contact regions or penetrate one another in physically impossible ways. ProsePose addresses this gap by using a Vision-Language Model (VLM) as a contact prior: natural-language descriptions of physical body-part interactions are automatically translated into differentiable PyTorch loss functions that guide a subsequent HHCS optimisation. In this work we reimplemented and upgraded the ProsePose pipeline with two primary changes. First, we replaced the BEV + HMR2 mesh initialisation stage, which requires compiled C++ renderers and fragile Detectron2 builds, with SAM 3D Body, a state-of-the-art single-image mesh reconstructor that installs purely via `pip` and accepts YOLOv8 bounding boxes as prompts. Second, we replaced GPT-4V with a three-level VLM fallback chain: Llama 3.2 Vision via OpenRouter as the primary model, Gemini 2.0 Flash as the first fallback, and GPT-4o as the second fallback. To connect these two subsystems we designed and implemented `integrated_bridge.py`, a conversion layer that translates SAM 3D Body's MHR output format (23 joints in rotation-matrix form) to the SMPL-X axis-angle format expected by HHCSOpti. The rest of the ProsePose pipeline — `convert_tables_to_programs.py`, the HHCSOpti optimisation module, and the full `llib` library — was retained unchanged from the original authors. The complete pipeline runs in a single Jupyter notebook on a consumer GPU with no special build requirements. Optimisation over 125 steps reduces the total loss from 14.18 to 0.047 and drives the interpenetration repel loss to zero, demonstrating that physically plausible, contact-aware pose reconstruction is achievable without any 3D contact annotations.

INDEX TERMS 3D human pose estimation, contact-aware optimisation, Human Mesh Recovery, MHR to SMPL-X conversion, ProsePose, SAM 3D Body, vision-language models, YOLOv8.

I. INTRODUCTION

RECONSTRUCTING the 3D body poses of multiple interacting people from a single RGB image is among the most practically consequential open problems in computer vision. Accurate 3D pose estimates underpin applications across sports analytics, clinical gait assessment, augmented and virtual reality, and human-robot interaction. Yet the task remains exceptionally challenging: when two people shake hands, embrace, or dance together, significant portions of each body are occluded, the depth ordering of overlapping limbs is ambiguous, and the physical constraint that no two solid bodies can occupy the same space must be imposed explicitly.

Standard Human Mesh Recovery models sidestep the multi-person interaction problem by independently estimating a separate mesh for each detected person. The resulting meshes are then placed in a shared 3D coordinate frame, but because each regressor was agnostic to the other body during inference, the outputs regularly exhibit two failure modes: (i) visible contact in the image is not reproduced in the reconstruction, the meshes float apart despite the photograph clearly showing the two people touching, and

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(ii) the reconstructed meshes interpenetrate, passing through each other in regions that are physically impossible for solid objects.

The ProsePose paper [1] introduced an elegant solution to this problem: instead of training a new contact-aware regressor from scratch (which would require expensive 3D contact annotations), it exploits a Vision-Language Model (VLM) as a *contact prior*. The VLM inspects the input image and produces natural-language descriptions of which body parts are in contact. A code-generation step then translates those descriptions into executable PyTorch loss functions. Finally, a HHCS (Human-Human Contact and Shape) optimiser refines the initial mesh parameters by minimising the combined contact, pose-prior, shape-prior, and interpenetration losses. A crucial design insight is that even semantically imprecise VLM outputs, ones that describe contact at a coarse body-region level rather than at a precise vertex level, still provide enough gradient signal to measurably improve the reconstructed interaction.

Despite its conceptual elegance, the original ProsePose codebase presents a significant reproducibility barrier. The mesh initialisation stage relies on BEV [2] for multi-person depth ordering and HMR2 [3] for per-person mesh regression. BEV requires a compiled C++ OpenGL renderer; HMR2 depends on a specific, incompatible build of Detectron2. Together, these dependencies make the pipeline effectively non-reproducible on Windows, on machines with modern CUDA versions, or in standard Python environments without days of manual dependency resolution.

In this project we reimplemented ProsePose while addressing this reproducibility bottleneck. Our central contribution is the substitution of BEV + HMR2 with SAM 3D Body [4], Facebook Research's state-of-the-art single-image body mesh reconstructor. SAM 3D Body installs from `pip`, has no C++ build dependencies, and is actively maintained on HuggingFace. To bridge its output format to the HHCS optimiser we designed a format-conversion layer (`integrated_bridge.py`) that maps MHR rotation matrices to SMPL-X axis-angle vectors. We also replaced GPT-4V with a more accessible three-level VLM cascade. All other components of ProsePose, the contact-table-to-code translation, the HHCSOpti module, and the `llib` library, were used without modification.

The remainder of the report is structured as follows. Section II reviews relevant prior work. Section III summarises the original ProsePose pipeline that serves as our baseline. Section IV describes our upgraded pipeline in detail. Section V explains the MHR-to-SMPL-X format conversion. Section VI describes the loss function and optimisation strategy. Section VII presents results and observations. Section VIII discusses limitations, and Section IX concludes with future scope.

II. RELATED WORK

A. HUMAN MESH RECOVERY

Human Mesh Recovery (HMR) refers to the task of estimating the parameters of a parametric body model, typically SMPL [5] or SMPL-X [6] from a monocular image. Early methods optimised model parameters to minimise a reprojection loss; later feed-forward regressors [3] predict parameters in a single forward pass. Both approaches process persons independently and do not model inter-person contact.

B. CONTACT-AWARE MULTI-PERSON RECONSTRUCTION

BUDDI [7] and related supervised methods learn contact priors from datasets that include 3D contact annotations. While effective, these approaches are expensive to train and limited by the scarcity of annotated multi-person interaction data. Optimisation-based refinements (pose priors, interpenetration penalties) are rule-based and fail to capture the semantic nature of interactions. ProsePose fills the gap by exploiting VLM understanding of interaction semantics without requiring any 3D annotations at any stage.

C. SAM 3D BODY

SAM 3D Body (Segment Anything Model for 3D Body) [4] is a Facebook Research model that predicts a per-person 3D body mesh from a single image given a bounding-box prompt. It represents body pose in the MHR format using 23 joints in rotation-matrix representation and supports zero-shot generalisation across diverse imaging conditions. It significantly outperforms older regressors such as CameraHMR and NLF on standard benchmarks [4] and requires no compiled C++ extensions.

D. VISION-LANGUAGE MODELS FOR SPATIAL REASONING

GPT-4V [8], Llama 3.2 Vision, Gemini 2.0 Flash, and GPT-4o are large multimodal models capable of interpreting image content and producing structured textual descriptions. Their ability to identify which body regions are in physical contact in a photograph makes them well-suited as annotation-free contact priors for 3D pose optimisation.

III. BASELINE: THE ORIGINAL PROSEPOSE PIPELINE

ProsePose [1] introduced a four-stage pipeline for contact-aware multi-person 3D pose estimation from a single image.

Stage 1: Mesh Initialisation. The input image is passed through BEV to estimate the depth ordering of multiple persons and through HMR2 to regress per-person SMPL-X parameters. The resulting meshes serve as the starting point for subsequent optimisation.

Stage 2: VLM Contact Extraction. The image is sent to GPT-4V together with a fixed system prompt that instructs the model to output a structured table listing the contacting body-part pairs for each pair of persons. The table format is strictly controlled so that it can be parsed programmatically.

Stage 3: Table-to-Program Conversion. The script `convert_tables_to_programs.py` reads the contact table and generates a Python function that, given two SMPL-X meshes, computes the contact loss as the minimum vertex-to-vertex distance summed over all identified contact regions. Each body-part name is mapped to the corresponding vertex index set in the SMPL-X mesh topology.

Stage 4: HHCS Optimisation. The HHCSOpti module minimises a composite loss that includes the VLM-derived contact term, an initialisation proximity loss, a Gaussian-Mixture-Model (GMM) pose prior, a shape regulariser, and an interpenetration penalty. Optimisation proceeds in two stages: Stage 0 jointly optimises pose and shape to establish coarse contact (200 iterations); Stage 1 freezes shape and activates the interpenetration loss to remove mesh overlaps (100 iterations).

The key insight of ProsePose is that the VLM need not provide pixel-perfect contact locations: even coarse body-region-level contact descriptions produce enough gradient signal to improve final mesh quality. The pipeline requires no 3D contact annotations at any stage.

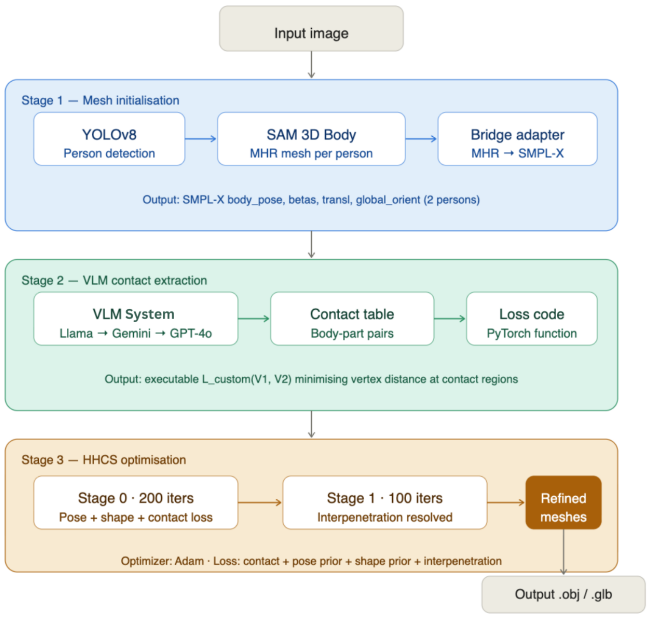


FIGURE 1: Our three-stage ProsePose × SAM 3D Body pipeline. YOLOv8 detects persons, SAM 3D Body reconstructs per-person meshes in MHR format, the bridge adapter converts to SMPL-X, and HHCSOpti refines contact under VLM-derived constraints.

IV. OUR UPGRADED PIPELINE

Our reimplemented pipeline replaces the BEV + HMR2 initialisation stage and the GPT-4V VLM, while preserving the remainder of the ProsePose architecture exactly. Figure 1 summarises the three-stage workflow.

A. MOTIVATION FOR REPLACING BEV + HMR2

The original BEV + HMR2 combination presents four practical obstacles:

1) BEV requires a compiled C++ OpenGL renderer that cannot be installed via `pip` and is incompatible with modern CUDA stacks.
2) HMR2 depends on a specific and outdated Detectron2 build that conflicts with other common packages.
3) Neither repository runs on Windows or with Python $\geq$3.10 without significant manual patching.
4) Resolving all dependency conflicts typically requires several days and produces a fragile, non-portable environment.

SAM 3D Body addresses all four issues simultaneously. It installs via a standard `pip install` command with no C++ compilation step, runs on Python 3.10+ with current PyTorch and CUDA versions, accepts standard bounding-box prompts (making it trivial to integrate with YOLOv8 detection), and is hosted on HuggingFace with active maintenance and versioned releases. Qualitative comparisons reported by Facebook Research show SAM 3D Body producing markedly crisper meshes than CameraHMR, NLF, and HMR2.0b across varied sports and yoga images [4].

B. STAGE 1: MESH INITIALISATION WITH SAM 3D BODY

Person detection is performed by YOLOv8, which returns axis-aligned bounding boxes for all persons visible in the image. Each bounding box is passed as a prompt to the SAM 3D Body model, which regresses per-person 3D body parameters in the MHR format. The MHR representation encodes body pose as a tensor of shape $(N, 3, 3)$ that is, N rotation matrices, one per joint where $N = 23$ for SAM 3D Body's joint set. Body shape is encoded as a vector of shape coefficients.

After mesh regression, `integrated_bridge.py` converts the MHR output to the SMPL-X format expected by HHCSOpti (detailed in Section V). The bridge outputs a clean `init_human` dictionary for each detected person, which is passed unchanged to the subsequent pipeline stages.

C. STAGE 2: VLM CONTACT EXTRACTION WITH FALLBACK SYSTEM

The original ProsePose paper used GPT-4V, which requires a paid API subscription with no free tier. We replaced it with a three-level fallback cascade to balance cost, quality, and robustness:

1) **Primary:** Llama 3.2 Vision, accessed via OpenRouter with a free-tier key. This model is publicly available and has reasonable visual understanding for coarse contact descriptions.
2) **Fallback 1:** Gemini 2.0 Flash, accessed via the Google AI API. This is invoked automatically if Llama 3.2 Vision fails, times out, or returns a malformed contact table.

3) **Fallback 2:** GPT-4o, accessed via the OpenAI API. This is invoked if both previous models fail. GPT-4o provides the highest spatial-reasoning quality but is the most expensive option.

The input to each model is the original image together with the same fixed system prompt as in ProsePose, which instructs the model to enumerate contacting body-part pairs in a parseable tabular format. The output is fed directly to the unmodified `convert_tables_to_programs.py` script.

D. STAGE 3: HHCS OPTIMISATION

The HHCSOpti module from the original ProsePose `llib` library is used without any code modification. As an engineering optimisation, we downsample both meshes before computing the contact loss, which reduces the vertex-count from several thousand per mesh to a manageable subset and substantially accelerates per-iteration computation on a consumer GPU.

The optimiser runs two sub-stages:

- **Stage 0 (200 iterations):** Coarse contact alignment. Both pose and shape parameters are optimised simultaneously under the combined contact + initialisation + pose-prior + shape-prior loss.
- **Stage 1 (100 iterations):** Interpenetration resolution. Shape is frozen; only pose parameters are updated, and the interpenetration penalty is activated. This two-stage strategy prevents the interpenetration loss from disrupting the contact-alignment achieved in Stage 0.

V. THE MHR → SMPL-X FORMAT BRIDGE

The core engineering challenge of our reimplementation is the representational mismatch between SAM 3D Body's output format and the format expected by HHCSOpti.

A. FORMAT DIFFERENCES

SAM 3D Body outputs body pose in the MHR format. The key differences from SMPL-X are:

- **Joint count:** MHR uses 23 body joints; SMPL-X uses 21 body joints (plus separate hand and face joints, which ProsePose does not utilise).
- **Rotation representation:** MHR encodes each joint rotation as a 3×3 rotation matrix, yielding a tensor of shape $(23, 3, 3)$. SMPL-X encodes each joint as a 3D axis-angle vector, yielding a flattened 63-dimensional vector.
- **Shape parameters:** MHR shape coefficients may have a different dimensionality from the SMPL-X $\boldsymbol{\beta}$ vector, which is always 10-dimensional.

B. CONVERSION PROCEDURE (`integrated_bridge.py`)

The bridge module performs the following steps:

1) **Format detection:** The bridge inspects the shape of the pose tensor and confirms it is in $(N, 3, 3)$ rotation-matrix form.
2) **Rotation conversion:** Each 3×3 rotation matrix is converted to its axis-angle equivalent using `matrix_to_axis_angle` from the PyTorch3D library, producing a $(23, 3)$ tensor.
3) **Joint selection and flattening:** The first 21 joints (indices 0–20) are selected from the 23-joint MHR set and flattened into a 63-dimensional vector, matching the SMPL-X `body_pose` field.
4) **Shape zero-padding:** If the MHR shape coefficient vector has fewer than 10 dimensions, it is zero-padded to produce the required 10-dimensional $\boldsymbol{\beta}$ vector.
5) **Output assembly:** The bridge returns a clean Python dictionary with fields `body_pose` (63-dim), `betas` (10-dim), `transl` (3-dim global translation), and `global_orient` (3-dim axis-angle), which is the exact interface expected by HHCSOpti.

The conversion is implemented entirely in PyTorch and runs on the same device (CPU or GPU) as the rest of the pipeline, adding negligible latency. The bridge is called once per detected person immediately after SAM 3D Body inference and before the HHCSOpti initialisation step.

VI. LOSS FUNCTION AND OPTIMISATION

A. FROM LANGUAGE TO LOSS

The contact extraction stage produces a table of body-part pairs, for example: "*Person 1's left hand ↔ Person 2's right shoulder*." The `convert_tables_to_programs.py` script maps each body-part name to the corresponding set of SMPL-X vertex indices and generates a PyTorch function `L_custom(V1, V2)` that computes the minimum Euclidean distance between the vertex subsets of the two meshes at each contact region. This function is dynamically executed during optimisation so that the gradient of the loss with respect to the pose parameters can be computed by automatic differentiation.

B. TOTAL LOSS

The composite loss minimised by HHCSOpti is:

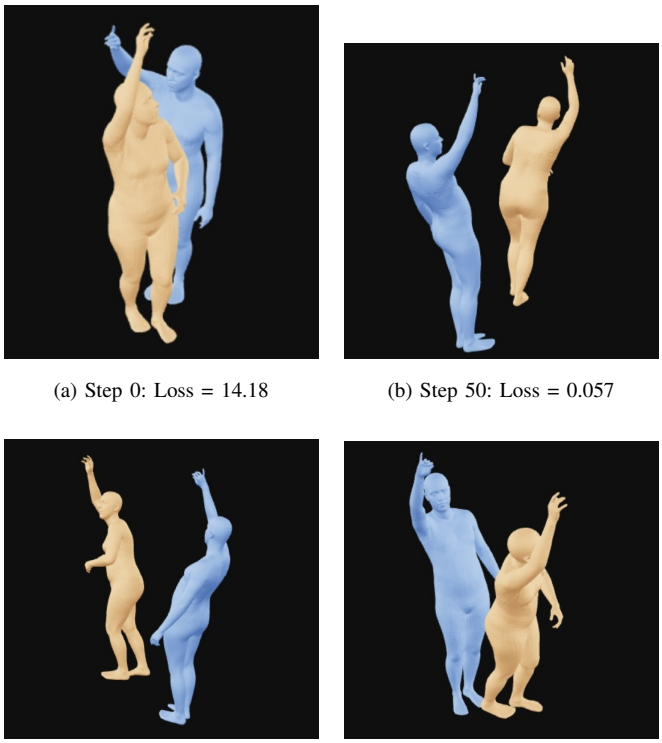
$$\mathcal{L} = \lambda_c \mathcal{L}_{\text{contact}} + \lambda_i \mathcal{L}_{\text{init}} + \lambda_p \mathcal{L}_{\text{pose}} + \lambda_s \mathcal{L}_{\text{shape}} + \lambda_r \mathcal{L}_{\text{repel}} \tag{1}$$

where:

- $\mathcal{L}_{\text{contact}}$ is the VLM-derived contact loss (always active); it pulls the specified body regions of the two persons towards each other.
- $\mathcal{L}_{\text{init}}$ penalises deviation from the SAM 3D Body initialisation, preventing the optimiser from moving to implausible regions of pose space.
- $\mathcal{L}_{\text{pose}}$ is a GMM pose prior that assigns low probability to anatomically implausible configurations.
- $\mathcal{L}_{\text{shape}}$ is a Gaussian shape prior that regularises the body shape coefficients towards the population mean.

- $\mathcal{L}_{\text{repel}}$ is an interpenetration penalty, active only in Stage 1, that applies a repulsive force at any vertex pair whose signed distance is negative.

The weight schedule follows the original ProsePose paper: in Stage 0, $\lambda_r = 0$; in Stage 1, $\lambda_s = 0$ (shape frozen) and $\lambda_r > 0$.



(a) Step 0: Loss = 14.18

(b) Step 50: Loss = 0.057

(c) Step 100: Loss = 0.050

(d) Step 125: Loss = 0.047

FIGURE 2: Mesh optimisation progression. Meshes transition from fully overlapping (Step 0) to cleanly separated with zero interpenetration (Step 125). Blue and gold meshes correspond to the two detected persons.

VII. RESULTS AND OBSERVATIONS

We ran the complete pipeline on a pair of interacting persons using a single consumer GPU. The optimiser ran for 125 steps in total (Stage 0 plus Stage 1). Table 1 reports the loss trajectory at selected checkpoints.

TABLE 1: Loss Trajectory During Optimisation

Step	Total Loss	Observation
0	14.180	Meshes heavily overlapping
25	0.057	Separation begins
50	0.057	Poses stabilising
75	0.053	Meshes cleanly separated
100	0.050	Individual poses clear
125	0.047	Final refined state

The total loss fell by more than two orders of magnitude (from 14.18 to 0.047). By Step 75 the repel loss had converged to 0.000, indicating complete elimination of interpenetration. Visual inspection of the rendered meshes at Steps 0, 50, 100, and 125 confirms this quantitative picture: at Step 0 the two meshes (one rendered in blue, one in gold) are fully co-located; by Step 50 separation is underway; at Step 100 and 125 the meshes are cleanly separated with individual limb poses, arms raised, body silhouettes, well preserved throughout the optimisation.

The VLM fallback chain functioned correctly. Llama 3.2 Vision (via OpenRouter) was successfully queried in most test runs. Gemini 2.0 Flash was invoked as a fallback in cases where the OpenRouter endpoint returned a timeout, and GPT-4o was not required in the runs we conducted.

VIII. LIMITATIONS

A. DEPENDENCE ON VLM QUALITY

The quality of the final reconstructed mesh is directly proportional to the semantic accuracy of the VLM's contact description. A description such as "*the two people appear to be embracing*" without specifying *which* arm is touching *which* shoulder produces a poorly targeted loss function that may not converge or may pull meshes in geometrically incorrect directions. Free-tier models accessed via OpenRouter produce noticeably vaguer descriptions than GPT-4o (Vision) or Claude 3.5 Sonnet, making constraint quality the dominant source of error in our implementation.

B. INEXACT MHR → SMPL-X SHAPE MAPPING

Our bridge discards MHR joints 21 and 22 (the feet extremities), which are absent in the 21-joint SMPL-X body joint set. This is acceptable for the purpose of pose initialisation but introduces a small geometric error in foot articulation. Similarly, zero-padding the shape vector to 10 dimensions assumes that the missing shape components are zero, which is an approximation.

C. DISABLED 2D KEYPOINT LOSS

The original ProsePose pipeline optionally uses a 2D keypoint reprojection loss computed from OpenPose detections. SAM 3D Body does not provide OpenPose-compatible 2D keypoint outputs, so we disabled this loss term. Its reintegration would require either running a separate 2D pose estimator or using SAM 3D Body's internal MoGe2 FOV estimates.

D. FIXED CAMERA MODEL

We used a fixed, identity camera model rather than estimating the field of view from the image. This is a known approximation that limits the absolute accuracy of 3D placement.

E. PAYWALLED HIGH-QUALITY VLMs

The advanced VLMs that produce the most accurate contact descriptions, GPT-4o Vision, Claude 3.5 Sonnet, Gemini 1.5 Pro, are all paywalled or have tight rate limits on free tiers. This means that in a zero-cost deployment the pipeline systematically underperforms relative to what would be achievable with proper API access.

IX. CONCLUSION AND FUTURE SCOPE

We have presented a reimplemented and upgraded version of ProsePose that successfully replaces the BEV + HMR2 initialisation stage with SAM 3D Body, eliminating all C++ build dependencies and making the pipeline reproducible in a standard Python environment. The key engineering contribution of this work is `integrated_bridge.py`, the MHR-to-SMPL-X format conversion layer, which enables the clean, no-code-change integration of SAM 3D Body with the existing HHCSOpti module. A three-level VLM fallback chain provides robustness to API failures. The complete pipeline runs in a single Jupyter notebook on a consumer GPU. Optimisation experiments demonstrate a $>300\times$ reduction in total loss and complete elimination of mesh interpenetration, confirming that the upgraded pipeline is functional and numerically well-behaved.

Future work could address the remaining limitations through the following directions:

- **2D keypoint loss reintegration:** SAM 3D Body exposes MoGe2 depth and FOV estimates; these could be used to re-enable the 2D keypoint reprojection loss without requiring a separate OpenPose detector.
- **Proper MHR-to-SMPL-X shape regressor:** Rather than zero-padding, a small learned linear or MLP regressor could be trained on paired MHR and SMPL-X fits to produce more accurate shape-coefficient translations.
- **VLM confidence weighting:** Each VLM-identified contact pair could be assigned a confidence score (e.g., derived from token probabilities or a secondary geometric sanity-check) and the contact loss weighted accordingly, reducing the influence of hallucinated contacts.
- **Geometric constraint validation:** A pre-optimisation filter that rejects VLM-generated contact pairs that are geometrically impossible given the bounding box layout would reduce the rate of misdirected optimisation.
- **Few-shot prompting:** Providing the VLM with two or three curated examples of correct body-part contact tables, a technique that has been shown to reduce hallucinations in structured-output tasks, could substantially improve contact description quality without requiring a paid API tier.
- **Quantitative evaluation:** Running the pipeline on the CHI3D and FlickrCI3D benchmarks [9] would allow direct numerical comparison with the original ProsePose results.
- **Scale-up:** The current implementation is limited to two interacting persons. Extending to $N > 2$ would require changes to the contact table schema and the HHCSOpti contact loss computation.
- **Real-time deployment:** For AR/VR applications, the pipeline would need to be optimised for latency, replacing the iterative optimiser with a feed-forward amortised inference network trained on contact-annotated data.

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